Kingdom of Saudi Arabia Ministry of Education
Prince Sattam bin Abdulaziz University
College of Science and Humanities - Al-Aflaj Department of Computer Science

المملكة العربية السعودية
وزارة التعليم
جامعة الأمير سطام بن عبد العزيز
كلية العلوم والدراسات الانسانية - الأفلاج
قسم علوم الحاسب الآلي

Graduation Project – Part 1

Task Tracking System (TTS)

No	Student Name
1	سعد عبد الله السهلي
2	ربيع وقيان الخضران
3	محمد معجب محمد الدوسري

Supervised by

Dr. Lisaed Mahdi Bin Ammar

Semester One– 2024/ 2025

Kingdom of Saudi Arabia Ministry of Education
Prince Sattam bin Abdulaziz University
College of Science and Humanities - Al-Aflaj Department of Computer Science

المملكة العربية السعودية
وزارة التعليم
جامعة الأمير سطام بن عبد العزيز
كلية العلوم والدراسات الانسانية - الأفلاج
قسم علوم الحاسب الآلي

Task Tracking System (TTS)

A Graduation Project Report Submitted to Sattam Bin Abdulaziz University for the Degree of Bachelor of Science in Computer Sciences

No	Student Name
1	سعد عبد الله السهلي
2	ربيع وقيان الخضران
3	محمد معجب محمد الدوسري

Supervised by

Dr. Lasad Mahdi Bin Ammar

Semester One– 2024/ 2025

Abstract

Task Tracking System (TTS) is a program designed to improve project management and help companies efficiently complete their tasks, the platform enables managers to add tasks, assign them to employees, and track performance through precise tools and detailed reports, with a user-friendly interface, **TTS** aims to enhance productivity, reduce errors, and achieve tangible successes in the workplace.

الخلاصة

TTS هو برنامج يهدف إلى تحسين إدارة المشاريع ومساعدة الشركات على إنجاز مهامها بكفاءة، يتتيح المنصة للمديرين إضافة المهام، توزيعها على الموظفين، وتتبع الأداء عبر أدوات دقيقة وتقارير تفصيلية، بواجهة مستخدم سهلة، تسعى **TTS** إلى تعزيز الإنتاجية، تقليل الأخطاء، وتحقيق نجاحات ملموسة في بيئة العمل.

Acknowledgment

In the beginning, we thank Allah Almighty For The completion and for reaching this Stage and ending it as it should beautifully, we would like to express our sincere thanks and appreciation to **Dr. Lasad Mahdi Bin Ammar** for her invaluable guidance, suggestions, and encouragement, as we did, we finished this project with her help, and we would like to thank all the group members for their cooperation, suggestions, and persistence until the end. Thanks to everyone.

في البداية نحمد الله تعالى على ما أتمناه، وعلى وصولنا إلى هذه المرحلة، واختتامها كما ينبغي، على أكمل وجه، ونود أن نعرب عن خالص شكرنا وتقديرنا الدكتور **د. لسعد مهدي بن عمار** على توجيهاته واقتراحاته وتشجيعه الذي لا يقدر بثمن، كما فعلنا، أنهينا هذا المشروع بمساعدته، ونود أن نشكر جميع أعضاء المجموعة على تعاونهم واقتراحاتهم واستمرارهم حتى النهاية، شكرًا للجميع.

Table of Contents

Abstract **3**
الخلاصة **3**
Acknowledgment **4**
Table of Contents **5**
General Introduction **8**
Chapter 1 **9**
1.1. Introduction 10
1.2. Objectives of TTS 10
1.3. Scope of TTS 10
1.4. Type of TTS 11
1.5. Overview of TTS System 12
1.5.1. Trello 12
1.5.2. Monday.com 13
1.5.3. ClickUp 13
1.5.4. Basecamp 14
1.5.5. Jira 15
1.6. Comparison of TTS System 17
1.7. Methodology 18
1.8. Conclusion 19
Chapter 2 **20**
2.1. Introduction 21
2.2. Requirement Analysis 21
2.2.1. Functional Requirements 21
2.2.2. Non-Functional Requirements 22
2.3. Stakeholder 23
2.4. UML 24
2.5. System Use Case 24
2.6. Sequence Diagram 25
2.7. Class Diagram 28
2.8. Conclusion 30
Chapter 3 **31**
3.1. Introduction 32
3.2. Tools Selection: Database and programming 32
3.3. DATABASE 33
3.4. Django 37
3.5. User Interface Design 38
3.6. General Conclusion 41

3.7. Future Work 42
References **43**

General Introduction

In today's rapidly evolving business landscape, especially within the IT sector, effective project management is essential to achieving success and staying competitive. In this context,

The **Task Tracking System** (**TTS**) is a comprehensive tool designed to streamline project management and enhance team performance tracking.

TTS helps organizations to structure work, monitor progress, and prioritize tasks efficiently, thereby improving productivity and reducing wasted time and effort.

Organizations face challenges in task management and collaboration, causing delays and reduced productivity.

The proposed **TTS** offers a comprehensive solution with features like task assignment, progress tracking, and real-time collaboration to enhance efficiency.

Chapter 1
Introduction To Task Tracking System

1.1. Introduction

TTS is a software designed to streamline task management and enhance project efficiency for organizations. It addresses challenges such as clarifying expectations, tracking performance, and promoting data-driven decision-making. **TTS** helps businesses achieve their goals and adapt to future growth. It enhances transparency and accountability, driving long-term success.

TTS plays a crucial role in improving project management efficiency across various industries. It helps organizations overcome key challenges, such as unclear task expectations and poor performance tracking, by offering clear task assignments with deadlines and real-time monitoring of progress.

1.2. Objectives of TTS

TTS can have several objectives but the most common are:

- **Streamline Task Management**: Provide a software that allows managers to create, assign, and track tasks efficiently, ensuring clarity in responsibilities and deadlines.
- **Enhance Accountability**: Enable real-time monitoring of employee performance through progress tracking and time logging on tasks, fostering a proactive work culture.
- **Facilitate Data-Driven Decisions**: Generate detailed reports on team performance, offering insights for strategic planning and informed decision-making.

1.3. Scope of TTS

The scope of the TTS includes the following key functionalities and areas of focus:

- **Task Creation and Assignment**: Allows managers to create tasks, assign them to team members, and set deadlines, the system tracks task status and updates in real-time.
- **Time Logging and Progress Tracking**: Employees can log the hours spent on tasks, and the system provides real-time updates on progress, allowing managers to monitor performance efficiently.

- **Reporting and Analytics**: **TTS** generates detailed monthly and annual reports, summarizing completed tasks, overall team performance, and progress metrics, these reports assist in data-driven decision-making and project planning.
- **User-Friendly Interface**: **TTS** offers an intuitive interface, ensuring that teams can quickly adopt the software with minimal training, promoting smooth transitions and usage.
- **Security and Access Control**: Ensures that sensitive project data is securely stored and accessed only by authorized team members, with different roles and permissions configured in the system.

This scope is designed to address the core needs of project management while providing a flexible and scalable solution to organizations of various sizes.

1.4. Type of TTS

TTS systems can be mainly categorized into three types based on its deployment software:

- **Mobile Apps**: Designed for smartphones and tablets, allowing users to manage tasks on the go. The main features of such an apps includes push notifications for reminders, intuitive touch interface, and offline capabilities.
- **Desktop Applications**: Installed on personal computers, providing a robust environment for task and project management. This type of TTS usually contains advanced functionalities for detailed management, integration with other tools, and support for complex workflows.
- **Web application**: Accessible via web browsers, enabling task management from any device with internet connectivity. Common features includes real-time collaboration, easy updates and maintenance, and cross-software compatibility.

  Each **TTS** type offers unique advantages tailored to different user needs, ultimately enhancing task management and project tracking across various environments.

1.5. Overview of TTS System

Several TTS systems can be found in the literature. The following are among the most common ones.

1.5.1. Trello

Trello is a web-based project management application, similar to Kanban, developed by Trello Enterprise, a subsidiary of Atlassian. It was first established in 2011 by Software Creek Fog, which is now known as Glitch [1].

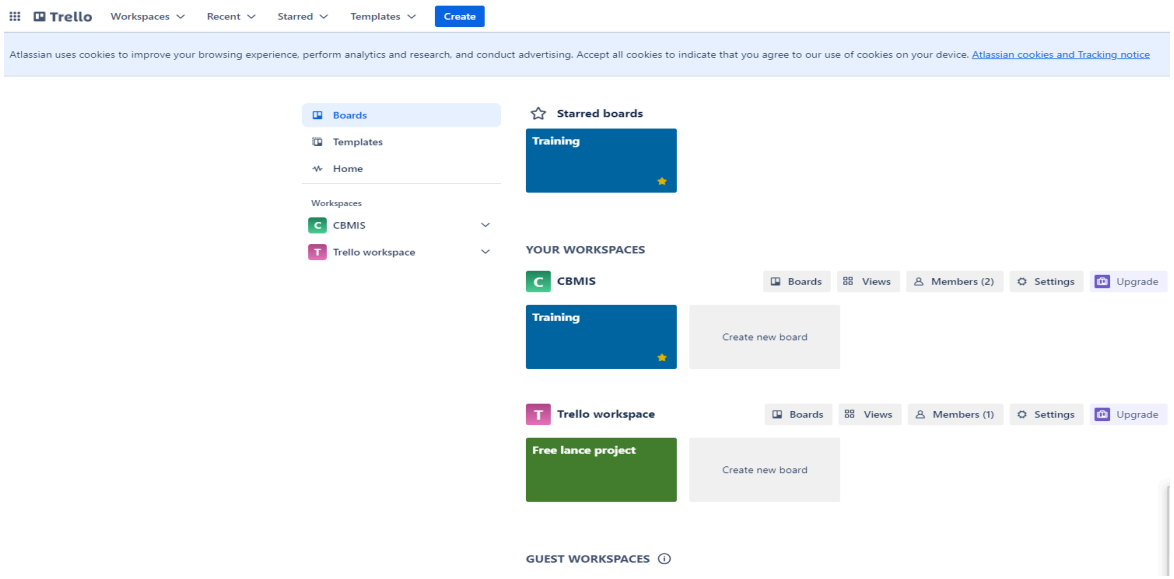


Figure 2 (Trello)

- **Advantages of Trello:**
  - Provides clients with updates on project progress and completed tasks.
  - Supports chat operations between team members.
  - Supports data encryption using SSL technologies.
- **Disadvantages of Trello:**
  - It only works when internet access is available.
  - Cards are limited to tasks within a single project in the free versions.
  - Storage space is limited.

1.5.2. Monday.com

Monday.com is an open software that enables individuals and teams to create the tools they need to manage every aspect of their work, the user-friendly visual interface includes various building blocks such as apps and integrations, allowing teams to smoothly build or customize their own solutions **[2]**.

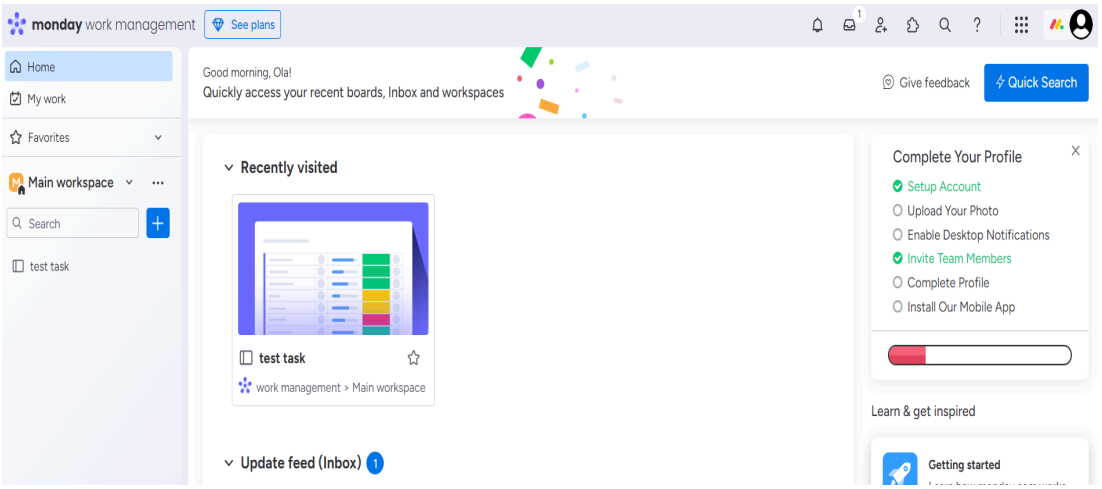


Figure 3 (Monday.com)

- **Advantages of Monday.com:**
  - AI support for implementing ideas.
  - AI tasks are executed on secure servers to maintain data security.
  - Mobile app support.
- **Disadvantages of Monday.com:**
  - Lack of support for customers in Arabic.
  - Does not support RTL (Right-to-Left) language formatting.
  - Difficulties in using the software with Arabic.

1.5.3. ClickUp

ClickUp is a versatile project management and productivity software that provides a comprehensive suite of tools to help teams plan, track, and collaborate on tasks, Its customizable features cater to various workflows and methodologies, making it suitable for teams of all sizes and industries **[3]**.

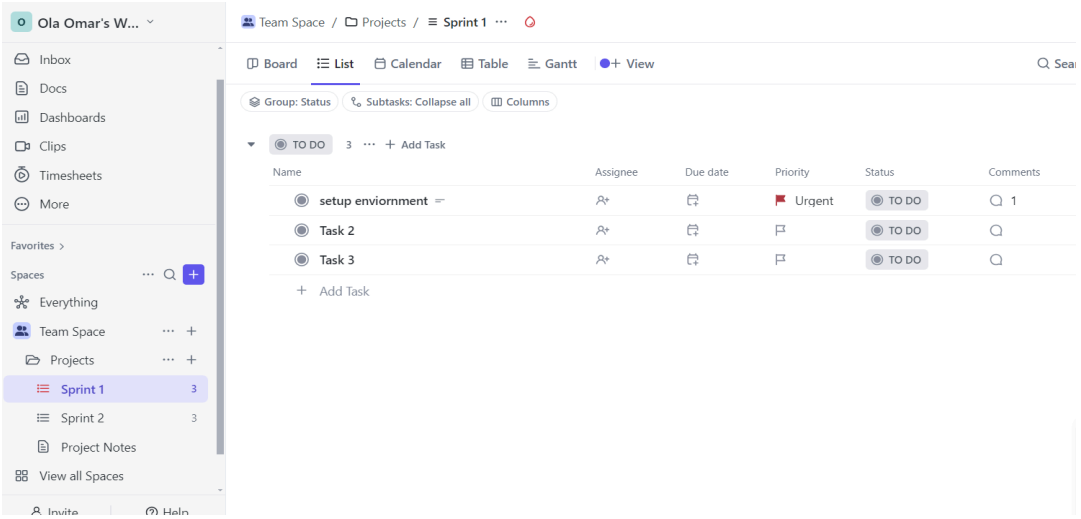


Figure 4 (ClickUp)

- **Advantages of ClickUp:**
  - **Customization:** Highly customizable dashboards, task statuses, and workflows to suit different team needs.
  - **All-in-One Solution:** Combines project management, time tracking, goal setting, and reporting in one software .
  - **Multiple Views:** Offers various views such as list, board, calendar, and Gantt chart to visualize tasks effectively.

- **Disadvantages of ClickUp:**
  - **Learning Curve:** The extensive features can be overwhelming for new users, requiring time to learn effectively.
  - **Performance Issues:** Some users report performance lags, especially with larger projects or teams.
  - **Notification Overload:** The system can generate a high volume of notifications, which may be distracting for users.

1.5.4. Basecamp

Basecamp is a project management software that offers a set of tools focused on reducing complexity and making project management more enjoyable and less routine, It is used by hundreds of thousands of teams to manage millions of projects **[4].**

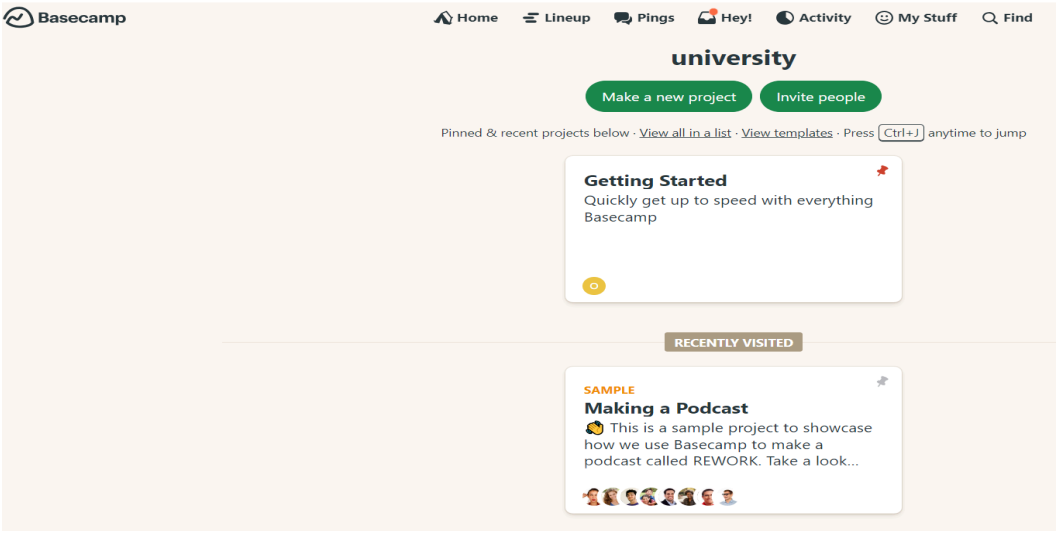


Figure 5 (Basecamp)

- **Advantages of Basecamp:**
  - Supports a mobile app version.
  - Supports a web version.
  - Simplifies project management.
- **Disadvantages of Basecamp:**
  - Does not support free trials.
  - Does not provide customer support in Arabic.
  - Does not support RTL (Right-to-Left) language formatting.

1.5.5. Jira

Jira is a software development tool used by engineers to track and manage tasks. You can use it to manage Agile and Waterfall projects, as well as to track bugs, features, and other work items. You can also configure **Jira** to work with many service management tools, making it a versatile task management tool **[5]**.

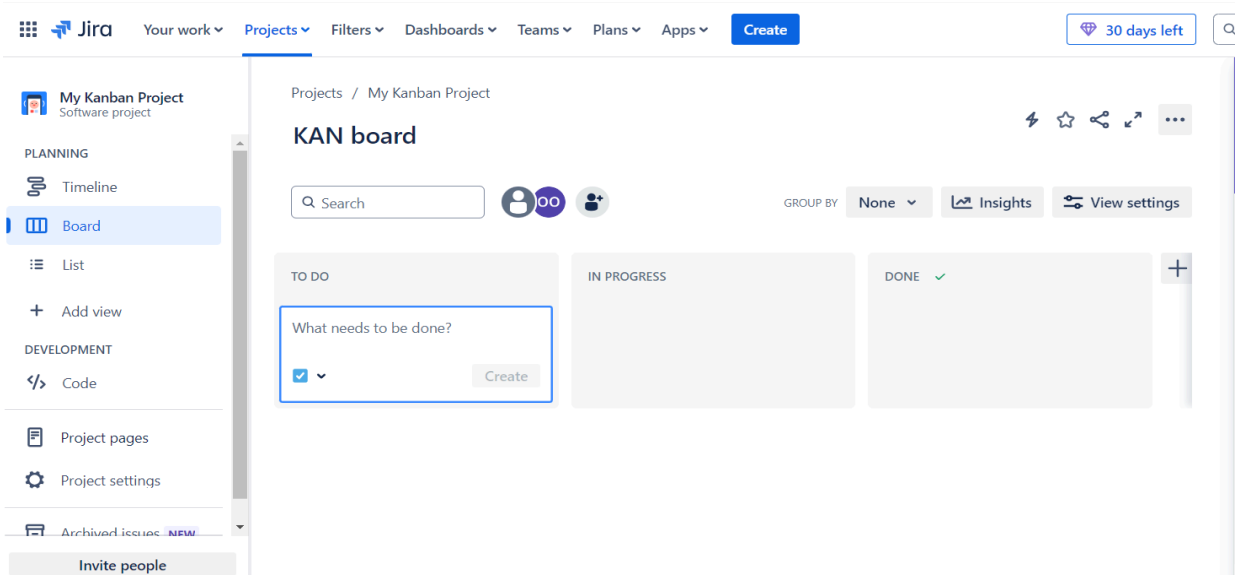


Figure 6 (Jira)

- **Advantages of Jira:**
    - **Free to Use:** Available at no cost for small teams (up to 10 users).
    - **Increased Productivity:** A flexible issue-tracking system and powerful reporting features.
    - **Better Workflow Visibility:** Automates processes and tracks progress in real-time.
- **Disadvantages of Jira:**
    - **Complex for Beginners:** The interface can be confusing for new users due to its size and abundance of features.
    - **Limited File Upload Size:** The maximum file upload limit of 10 MB.
    - **Lacks Communication Tools:** Jira does not include built-in communication features.

1.6. Comparison of TTS System

Features	TTS	Trello	Monday.com	Basecamp	ClickUp	Jira
Main Purpose	Task organization	Visual project mgmt	Work OS for tasks	Simplified project mgmt	Comprehensive project mgmt	Agile project mgmt
User- Friendly	User Friendly	Kanban boards	Modular & customizable	Clean & straightforward	Highly customizable	Complex
Task Management	Assign tasks, track progress	Cards across boards	Tasks with timelines	To-do lists	Multiple views & dependencies	Agile boards & sprints
Reporting	Yes	No	Yes	Yes	Yes	Yes
Price	Free	Free tier available	Free tier available	No	Free tier available	Free tier available
User Experience	Intuitive for basic needs	Easy for simple projects	Easy for simple projects	Easy for simple projects	Require training	Require training
Best For	All-in-one solution	Simple, visual tracking	Basic project mgmt	Basic project mgmt	All-in-one solution	All-in-one solution

Table 2.1: Comparison of Similar Mobile Applications

As you Table 2.1 although existing systems provide several advantages, they still suffer from some limitations **For example** complex user experience and limited file upload.

Hence, this project aims to organize projects and help companies to finish projects on time to overcome these limitations.

1.7. Methodology

To ensure effective project management and adaptability, we will follow the Agile methodology using the Scrum framework.

This approach is particularly suited for dynamic projects like the **TTS**, where flexibility, iterative progress, and teamwork are essential.

The Scrum Master will facilitate the team, coach them in Agile practices, and protect the team's focus by removing obstacles and maintaining alignment with Agile principles.

The choice of SCRUM can be justified by the fact that its iterative nature promotes responsiveness to changes and encourages continuous improvement, which is ideal for managing complex projects. It allows us to work in sprints, enabling frequent reassessment and adjustment of project priorities as necessary. Scrum also enhances team collaboration and accountability, essential for achieving project goals effectively. The following are the key concepts of the SCRUM methodology **[6]:**

- **Roles:** Scrum Master, Product Owner, and Development Team, each with distinct responsibilities.
- **Artifacts:** Product and sprint backlogs to track work, with increments marking progress.
- **Events:** Regular sprints, sprint planning, daily stand-ups, sprint reviews, and retrospectives.
- **Metrics:** Tools like burndown charts monitor progress, ensuring transparency and timely delivery of tasks.

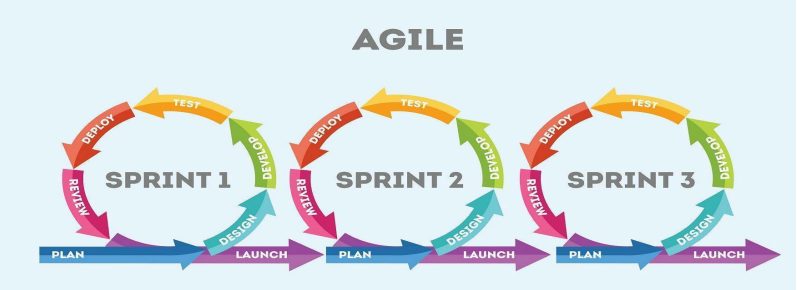


Figure 1 (Agile Scrum)

1.8. Conclusion

In this chapter, a comprehensive overview of the **TTS** project is provided. We present the key objectives, scopes and types of TTS. After that, a comparative study of the most common TTS software is provided. Finally, a brief description of the SCRUM methodology is provided. This methodology will be followed during this project to ensure its effectiveness.

Chapter 2

System Analysis and Design

2.1. Introduction

The purpose of this chapter is to conduct a comprehensive analysis of the **TTS** to understand its functional and non-functional requirements, identify key stakeholders, and presents the key UML diagrams allowing the design of our system.

2.2. Requirement Analysis

Requirement Analysis is a critical phase aimed at understanding and documenting what the system should do to meet user needs and business goals, It is divided into two main categories: Functional Requirements and Non-Functional Requirements.

2.2.1. Functional Requirements

Functional requirements define the core functionality that the system must provide to its users, For the **TTS**, the functional requirements include:

- **User Registration and Authentication**
  - Users must be able to create accounts with a unique username and password.
  - The system must allow registered users to log in and log out securely.
- **Role-Based Access Control**
  - The system must assign roles to users (Project Manager, Employee, Customer, Administrator) and control access based on roles.
  - Project Managers can create, assign, and monitor tasks.
  - Employees can update the status of tasks assigned to them.
  - Customers can review project progress and provide feedback.
- **Task Management**
  - Project Managers can create new tasks, set deadlines, and assign them to specific employees.
  - Employees can update task progress and record the time spent on each task.
  - The system must allow tasks to have different statuses (e.g., To-Do, In Progress, Completed).

- **Project Management**
    - Project Managers can create, manage, and archive projects.
    - The system must allow tracking of project progress with visual indicators (e.g., Gantt charts, Kanban boards).
- **Reporting and Analytics**
    - The system must generate performance reports for managers, showing task completion rates, time spent on tasks, and overall project progress.
    - The system must allow exporting reports in common formats (PDF, Excel).
- **Notifications and Alerts**
    - The system must send automatic notifications to users when a task is updated, a deadline approaches, or new tasks are assigned.
    - Users should be able to configure notification preferences.
- **Data Encryption and Security**
    - All sensitive data, including user credentials and project data, must be encrypted both at rest and in transit.

2.2.2. Non-Functional Requirements

Non-functional requirements describe the system's operational qualities and constraints. For **TTS**, these include:

- **Performance**
    - The system must handle concurrent users without significant degradation in performance.
    - Task updates and changes should be reflected in real-time with minimal delay.
- **Scalability**
    - The system must be able to scale as the number of users or projects increases without requiring major architectural changes.
- **Usability**
    - The interface should be intuitive and easy to navigate for users of all skill levels.

    - User training and onboarding materials should be provided to reduce the learning curve.
- **Availability**
    - The system must have 99.9% uptime and be available 24/7.
    - Scheduled maintenance should be communicated to users in advance.
- **Security**
    - Regular security audits must be performed to ensure data integrity and protection from unauthorized access.
- **Maintainability**
    - The system must be designed for easy updates and bug fixes, with minimal downtime.
    - Code must follow best practices to ensure that it is well-documented and easy to modify.

2.3. Stakeholder

The stakeholders of the **TTS** play crucial roles in ensuring its functionality and success.

The **Admin** is responsible for managing the system, user accounts, and overall functionality, focusing on system stability, user management, data security, and compliance with regulations. The **Customer** represents external clients or stakeholders who are invested in project outcomes and require transparency in project progress, the ability to provide feedback, and assurance that their needs are being met.

The **Project Manager** oversees project execution and team management, relying on efficient task management tools, clear progress visibility, and reliable performance monitoring.

Finally, the **Employee** works on tasks assigned by the project manager, updating task statuses and logging time spent to ensure accurate progress tracking and accountability.

2.4. UML

UML (Unified Modeling Language) is a tool or language used in the Design phase of software development, this language focuses on a collection of diagrams that provide visual representations of a system's structure and behavior, UML diagrams are divided into two main types:

- **Static Diagrams**: These diagrams focus on the system's structure and represent elements that remain constant over time For example Use Case Diagrams & Class Diagrams.
- **Dynamic Diagrams**: These diagrams focus on the system's behavior and the flow of activities or interactions over time For example Sequence Diagram.

2.5. System Use Case

The use case diagram is a behavior diagram, which is one of the diagrams in the Unified Modeling Language (UML) and it used to model the functional requirements of the system. A use case diagram shows the interaction between the system and the users of the system or any other system that use the proposed system.

The use case diagram for the **TTS** illustrates the interactions between various actors (Customer, Project Manager, Employee, and System Administrator) and the system's functionalities.
It highlights key actions such as user registration, task management, project tracking, and report generation. Each actor is linked to specific use cases, demonstrating their roles and responsibilities. This diagram aids in understanding system requirements and guides the design process.

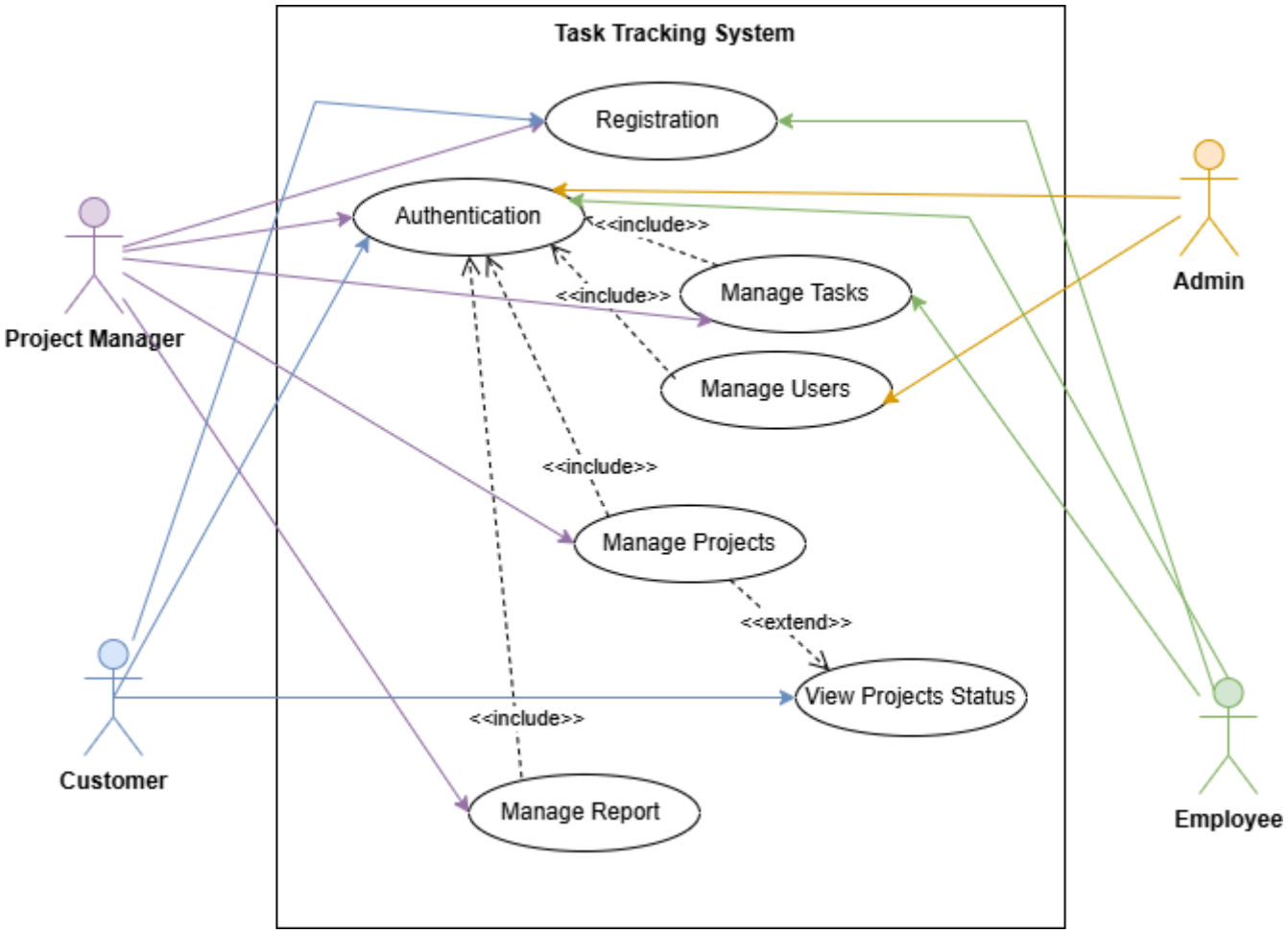


Figure 7 (Use Case Diagram) [7]

2.6. Sequence Diagram

This diagram shows the sequence of operations in the various functions of the system and the exchange of messages between parts of the system.

- **Project Manager Registration**



Figure 8 (System Use Case For Register as Project Manager) [8]

The sequence diagram outlines the registration process where a Project Manager submits details through the Registration Page, which are stored in the TTS Database. Based on data validation, the system either confirms successful account creation or returns an error for invalid or duplicate information.

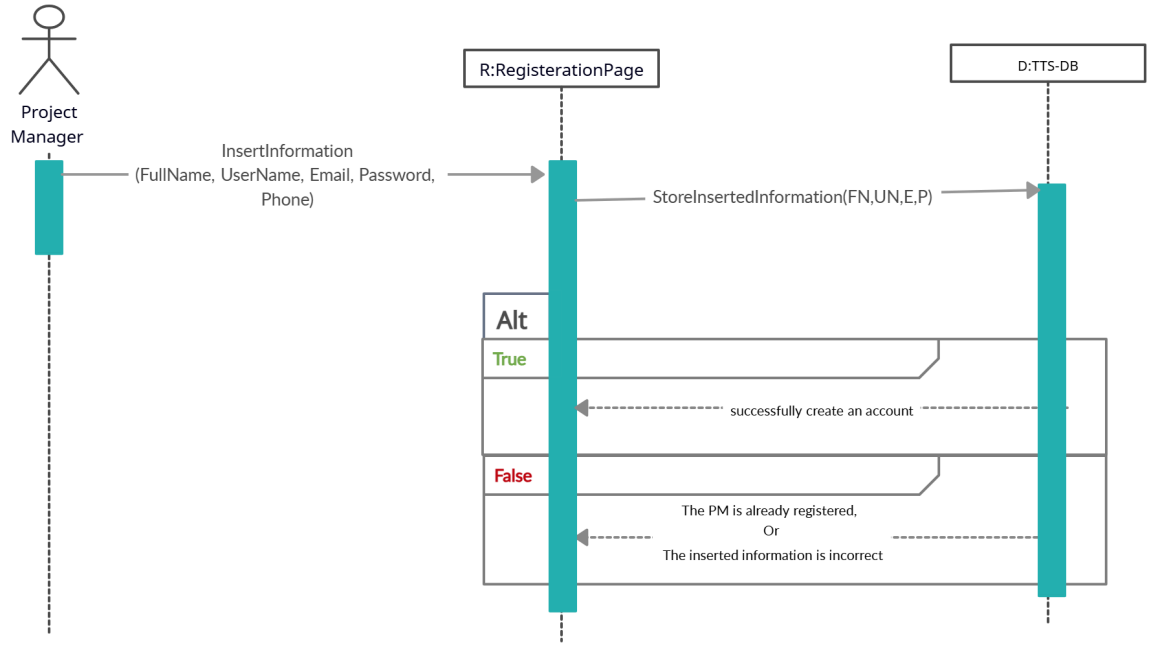


Figure 9 (Sequence Diagram for Register as Project Manager) [9]

- **Employee Authentication**

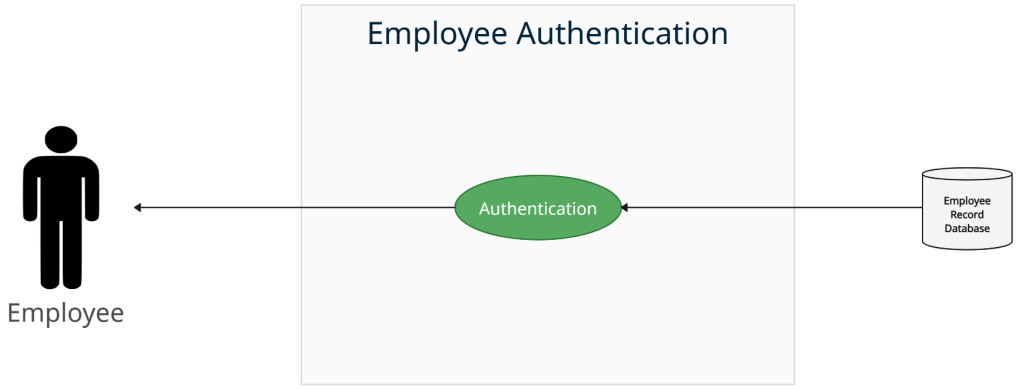


Figure 10 (System Use Case For Authentication as Employee) [10]

The Employee enters their credentials to log in, which the system validates; if successful, access to the dashboard is granted, allowing task performance until logout. Invalid credentials prompt an error message.

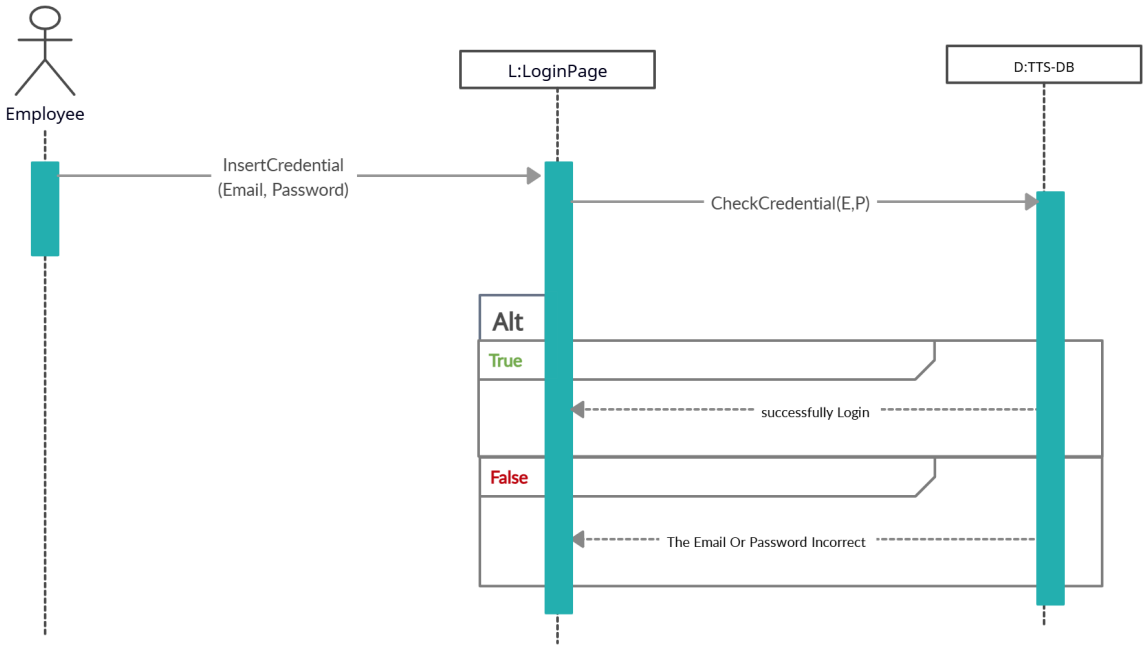


Figure 11 (Sequence Diagram for Authentication as Employee) [11]

- **Project Manager Manage Task (Add Task)**

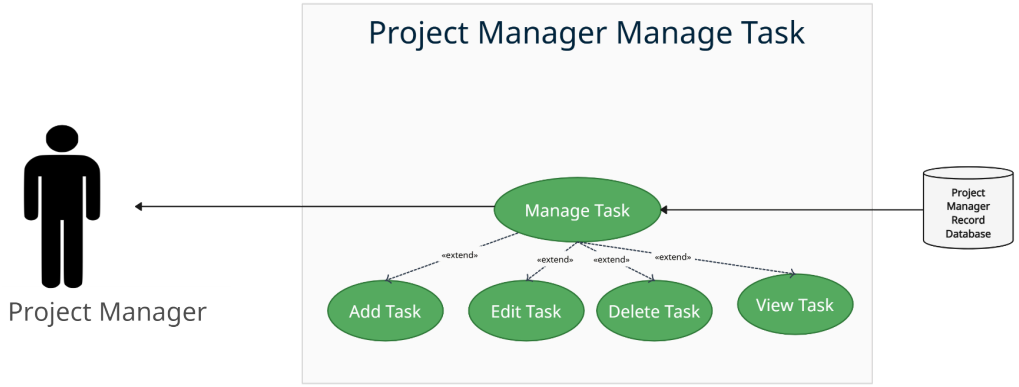


Figure 12 (System Use Case For Manage Task as Project Manager) [12]

The Project Manager accesses the task management feature to perform actions such as adding, editing, deleting, or viewing tasks. They input task details (e.g., name, deadline, status) for creation or modification, confirm changes, and the system updates the task database accordingly. Tasks can also be reviewed or removed as needed.

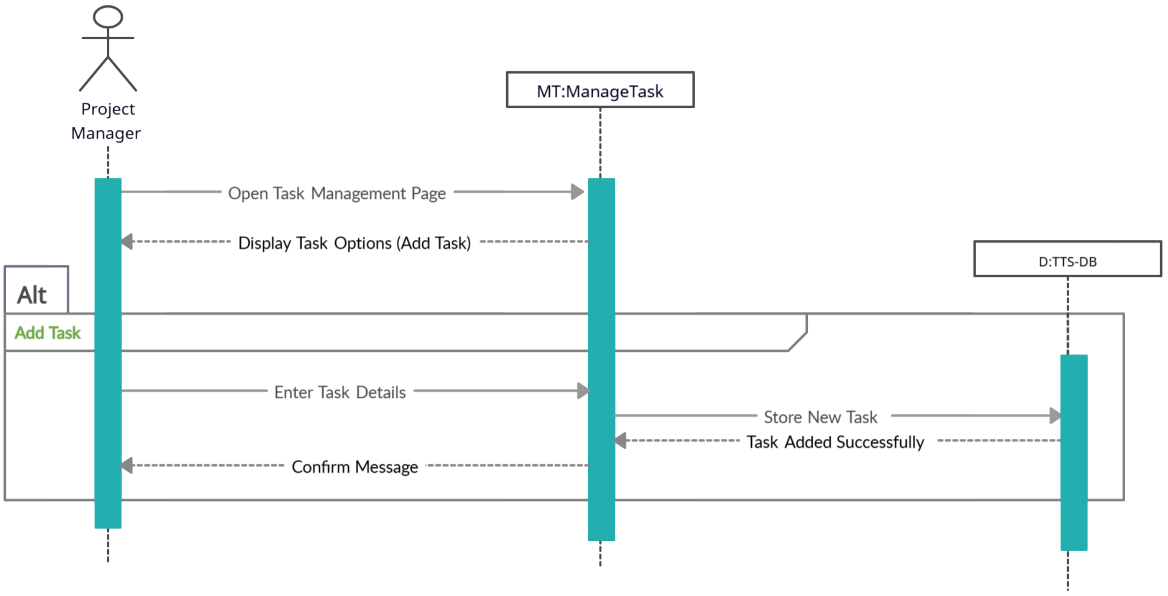


Figure 13 (Sequence Diagram for Manage Task as Project Manager) [13]

2.7. Class Diagram

The class diagram organizes the system's components and interactions, showing each user role's specific actions in the **TTS** system.

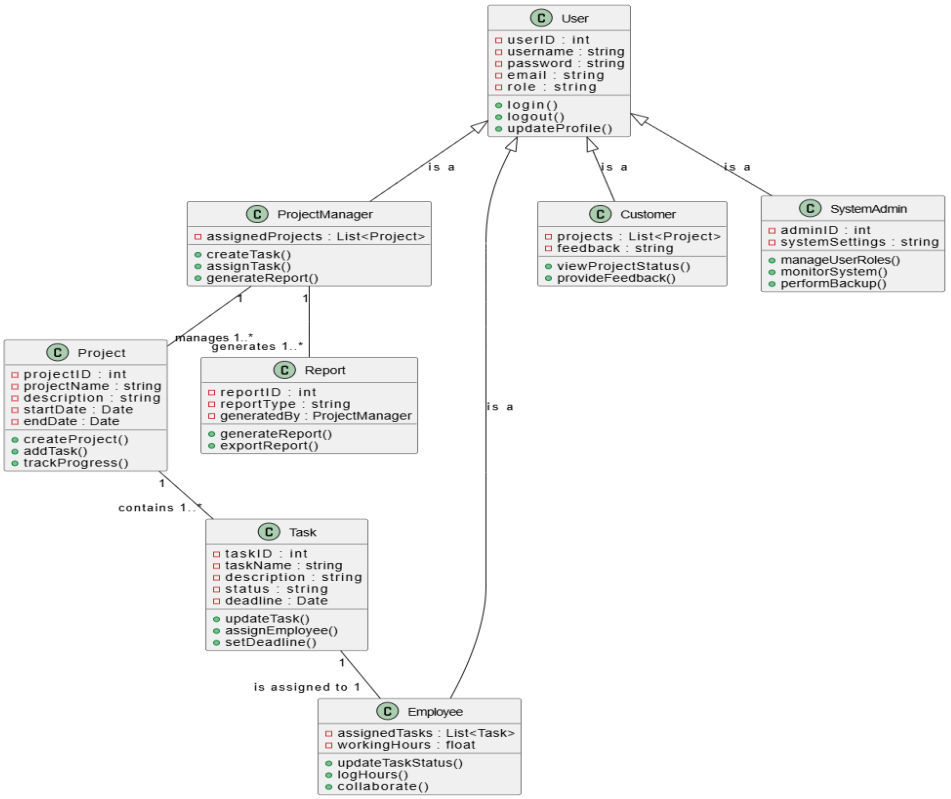


Figure 14 (Class Diagram) [14]

Below is an overview of the key classes involved in the **TTS**:

- **User**
    - **Attributes:** userID, username, password, email, role
    - **Methods:** login(), logout(), updateProfile()
- **ProjectManager (inherits from User)**
    - **Attributes:** assignedProjects
    - **Methods:** createTask(), assignTask(), generateReport()
- **Employee (inherits from User)**
    - **Attributes:** assignedTasks, workingHours
    - **Methods:** updateTaskStatus(), logHours(), collaborate()
- **Customer (inherits from User)**
    - **Attributes:** projects, feedback
    - **Methods:** viewProjectStatus(), provideFeedback()
- **Task**
    - **Attributes:** taskID, taskName, description, status, deadline
    - **Methods:** updateTask(), assignEmployee(), setDeadline()
- **Project**
    - **Attributes:** projectID, projectName, description, startDate, endDate
    - **Methods:** createProject(), addTask(), trackProgress()
- **Report**
    - **Attributes:** reportID, reportType, generatedBy

- **Methods:** generateReport(), exportReport()

- **Admin (inherits from User)**
  - **Attributes:** systemSettings
  - **Methods:** manageUserRoles(), monitorSystem(), performBackup()

The relationships between these classes will involve inheritance, composition, and associations. For example, a **ProjectManager** can manage multiple **Projects**, while a **Project** contains multiple **Tasks**. An **Employee** is assigned one or more **Tasks**.

2.8. Conclusion

In this chapter, we analyzed the core functionality and operational constraints of the **TTS** by outlining both functional and non-functional requirements. Additionally, we introduced the primary use cases that the system must support to ensure seamless project management. The insights gathered during this analysis phase will serve as the foundation for the design and implementation of the system, ensuring it meets the expectations of its users.

Chapter 3

System Implementation and Deployment Using Django Framework

3.1. Introduction

This chapter outlines the development environment and tools used to implement the Task Tracking System (TTS). It provides a detailed overview of the technologies chosen for development, including the database, programming framework, and user interface design. The aim is to justify tool selection, describe how components are integrated, and explain how the system was developed to meet functional and non-functional requirements.

3.2. Tools Selection: Database and programming

The selection of appropriate tools and technologies plays a crucial role in ensuring the system's efficiency, scalability, and maintainability. For the development of the Task Tracking System (TTS), the following tools were selected:

- **Programming Language & Framework**:
  The system is developed using **Python** with the **Django** framework. Django was chosen for its robust structure, rapid development capabilities, high security standards, and built-in administrative interface, which simplifies the management of system components.
- **Database**:
  - **MySQL** was selected as the database management system for handling structured data, user information, project records, and task logs. The **XAMPP** environment was used to facilitate easy local deployment and management of MySQL databases, offering a lightweight and user-friendly setup for development and testing.
- **Frontend Technologies**:

- **HTML**, **CSS**, **JavaScript**, and **Bootstrap** were used to design a responsive and user-friendly interface. These technologies ensure cross-browser compatibility and an interactive user experience across different devices.

- **Development Environment**:
  - **Visual Studio Code (VS Code)** was used as the primary Integrated Development Environment (IDE) due to its strong support for Python, a wide range of extensions, intuitive interface, and built-in debugging tools.

3.3. DATABASE

The Task Tracking System (TTS) relies on a relational database to manage and store data effectively. **MySQL** was selected as the database management system and integrated with Django through the XAMPP stack, providing a robust and scalable environment suitable for web applications.

Database Architecture

The system's database includes core application tables, user management tables (auto-generated by Django), and domain-specific entities. The schema is structured to support user roles, task assignment, project management, customer interaction, and reporting features.

Key Tables Description

Table Name	Description
customer	Stores customer data including contact information and login credentials.
company	Contains information about the companies working on or managing projects.
comment_task	Stores comments added by users or employees on specific tasks.

employee	Holds employee-specific details including department and assigned roles.
project	Contains information on each project such as timeline, status, and description.
project_manager	Stores data of project managers assigned to oversee project progress.
project_task	Tracks individual tasks under a project including deadlines and assignments.
report	Used to store generated reports for performance, progress, and time tracking.
users	Central table for all authenticated users of the system.
users_groups	Manages user group memberships (linked to Django's auth system).
users_user_permissions	Stores user-specific permissions.
authorized_group	Custom group roles for system-level access control.
authorized_group_permissions	Connects permissions to authorized groups. authorized_permissions Custom permissions for advanced role management.
django_admin_log	Logs admin actions for audit purposes.
django_content_type	Supports Django’s generic relationships and permission framework.
django_migrations	Tracks applied migrations to manage schema versions.
django_sessions	Manages user session data for authentication.

Table 3.1: Database Tables

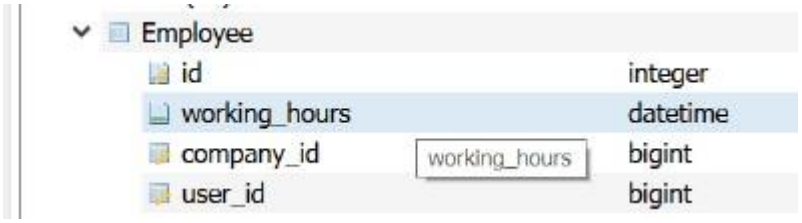


Figure 16 (Database Table Employee)

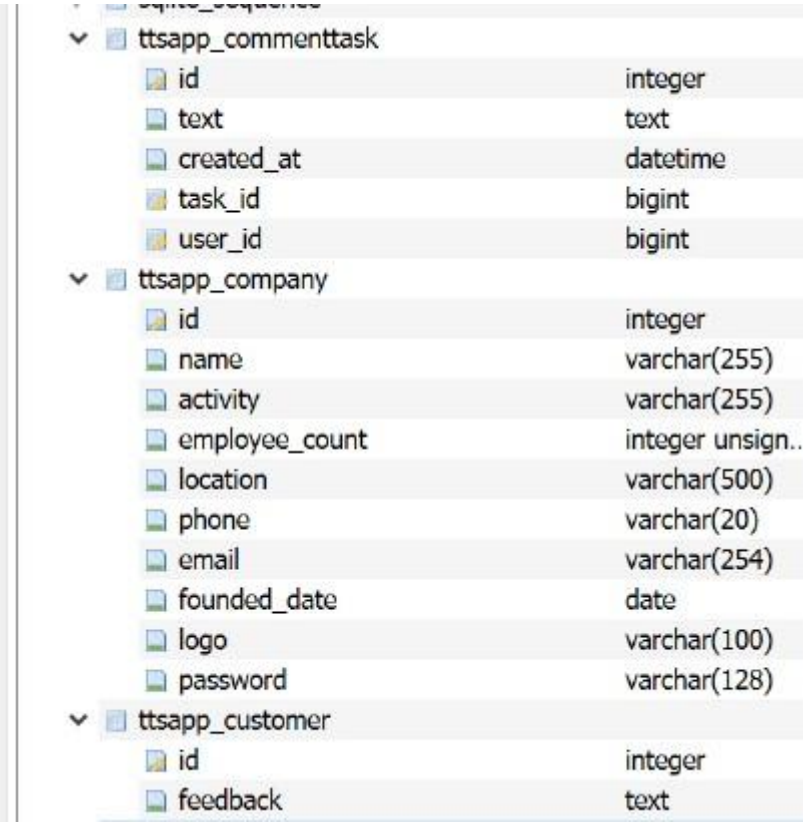
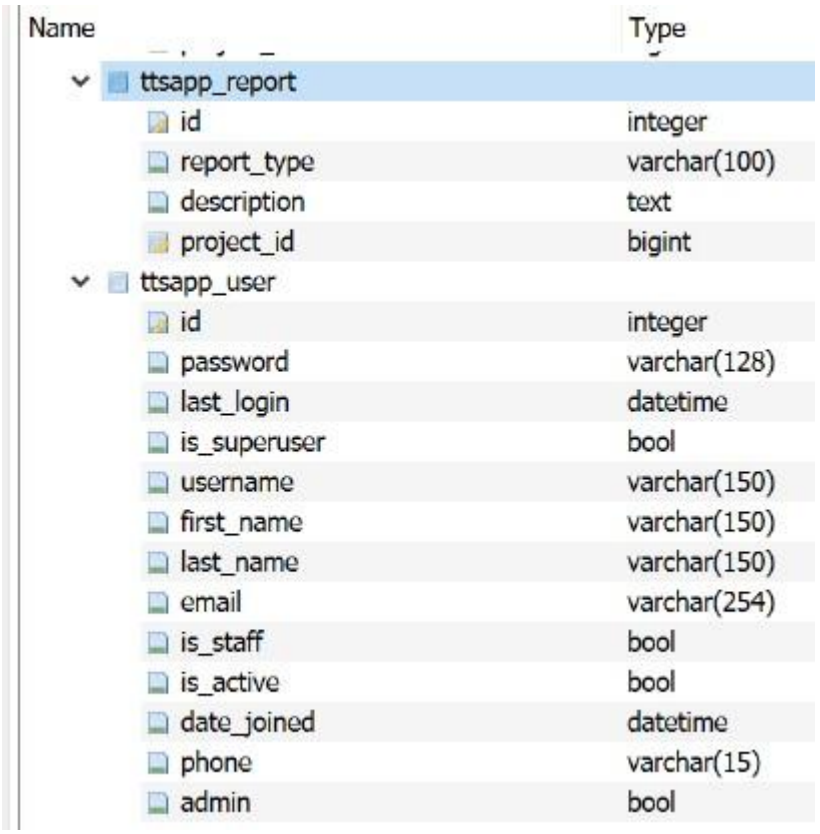
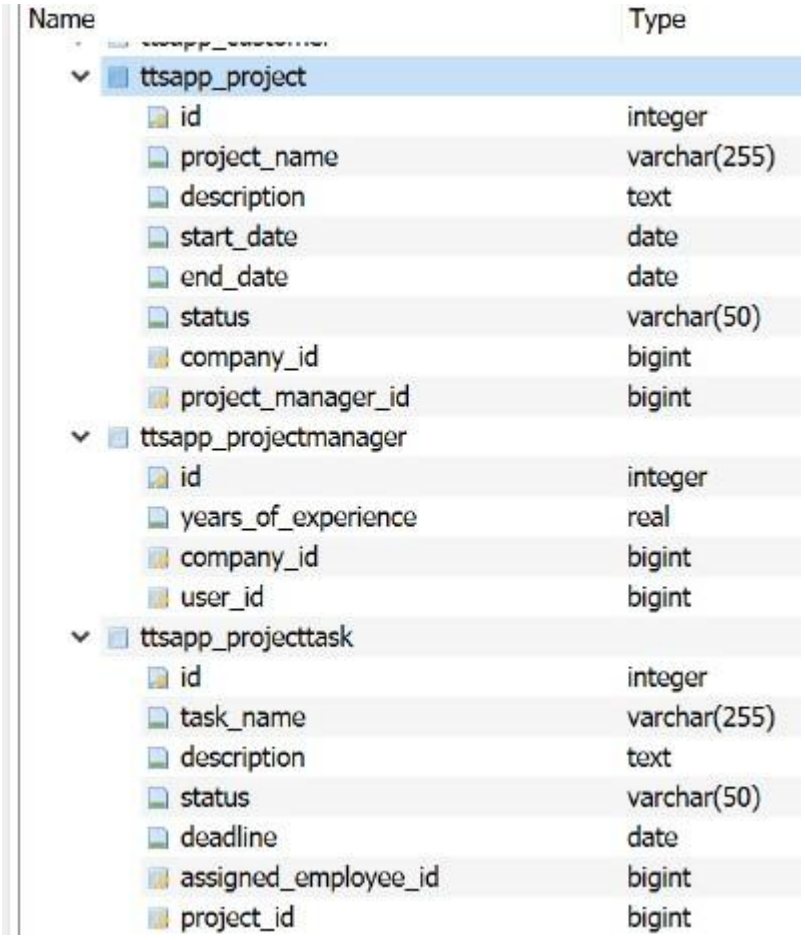


Figure 17 (Database Tables comment task, company, customer)



Name	Type
ttsapp_report	
id	integer
report_type	varchar(100)
description	text
project_id	bigint
ttsapp_user	
id	integer
password	varchar(128)
last_login	datetime
is_superuser	bool
username	varchar(150)
first_name	varchar(150)
last_name	varchar(150)
email	varchar(254)
is_staff	bool
is_active	bool
date_joined	datetime
phone	varchar(15)
admin	bool

Figure 18 (Database Tables report, user)



Name	Type
ttsapp_project	
id	integer
project_name	varchar(255)
description	text
start_date	date
end_date	date
status	varchar(50)
company_id	bigint
project_manager_id	bigint
ttsapp_projectmanager	
id	integer
years_of_experience	real
company_id	bigint
user_id	bigint
ttsapp_projecttask	
id	integer
task_name	varchar(255)
description	text
status	varchar(50)
deadline	date
assigned_employee_id	bigint
project_id	bigint

Figure 19 (Database Tables project, project manager, project task)

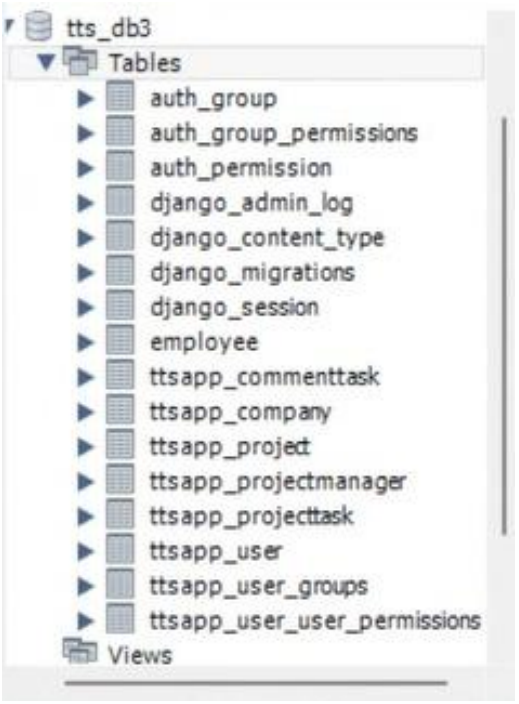


Figure 20 (Database All Tables)

Database Management Highlights

- **Relationships**: Foreign keys enforce consistency across users, projects, tasks, and feedback.
- **Normalization**: The design follows third normal form (3NF) to eliminate redundancy.
- **Security**: Sensitive data such as passwords are stored in hashed form.
- **Extendability**: The schema is built to support future features like file attachments, notifications, or enhanced reporting.

3.4. Django

Django, a high-level Python web framework, was selected for its simplicity, security features, and rapid development capabilities. It follows the MVC (Model-View-Controller) architecture, making it ideal for organizing system logic.

Benefits of using Django include:

- Built-in user authentication and authorization system.
- Automatic admin interface for data management.
- ORM (Object-Relational Mapping) for efficient database interactions.
- Middleware for security (e.g., CSRF protection, session management).
- Support for scalable application development.

3.5. User Interface Design

3.5.1. login Interface

This login interface provides secure access for three user roles: Company, Project Manager, and Employee, each user is directed to a customized dashboard based on their role after successful authentication.

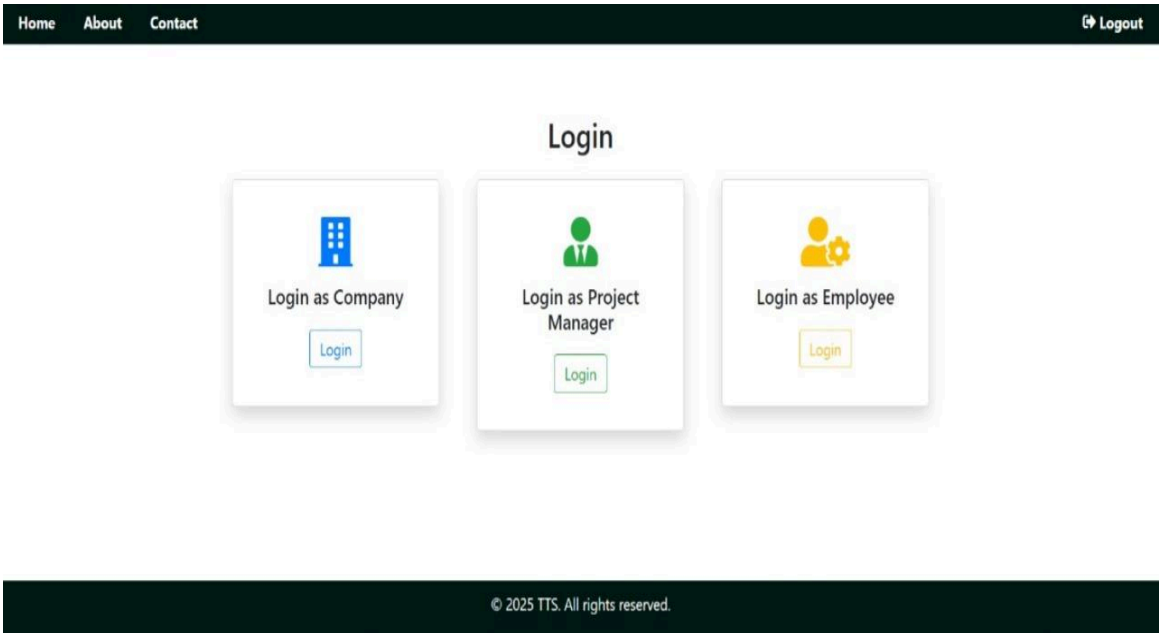


Figure 21 (Login Interface)

3.5.2. login As Company Interface

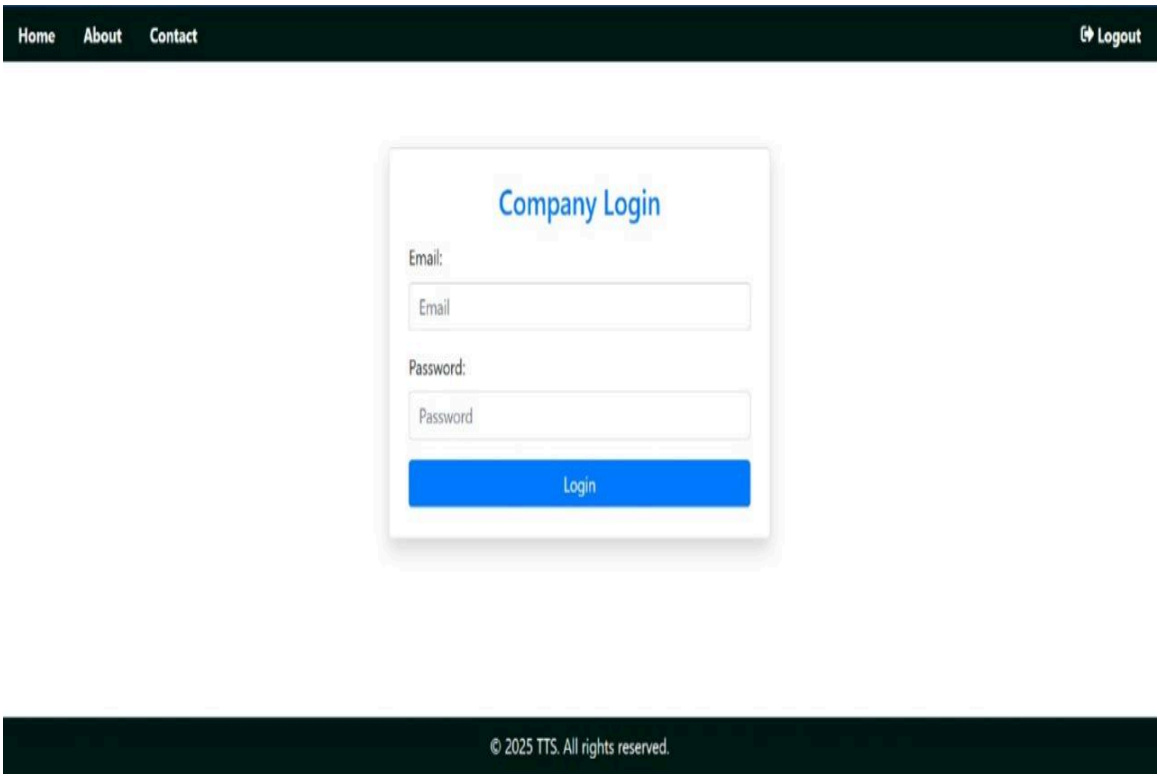


Figure 22 (Login As Company Interface)

3.5.3. View Task Interface

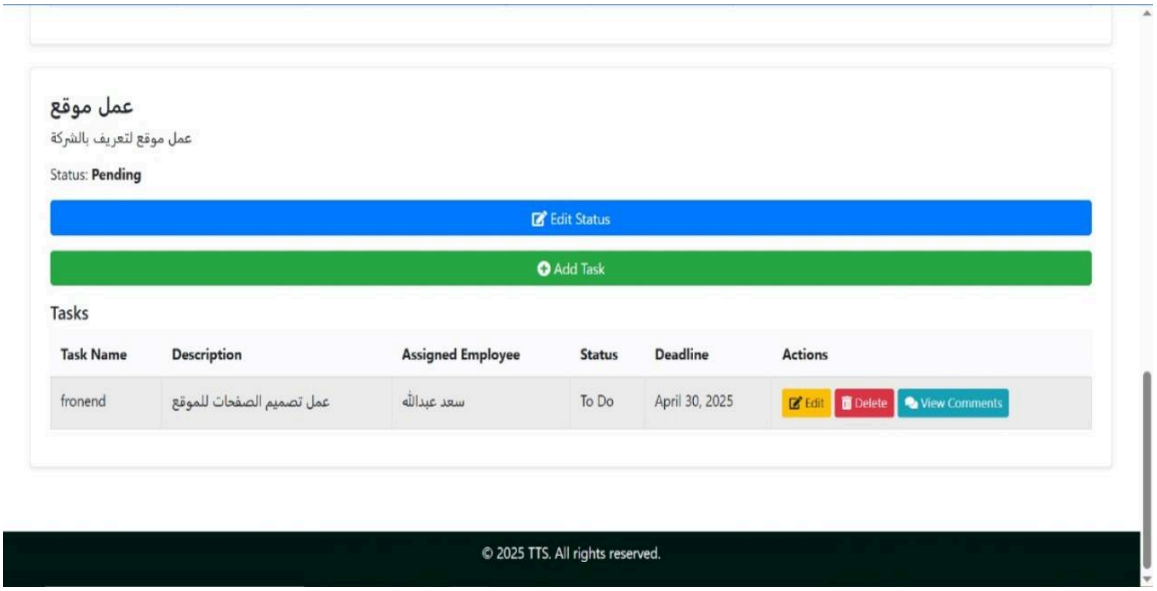


Figure 23 (View Task Interface)

3.5.4. Add Comment to Task Interface

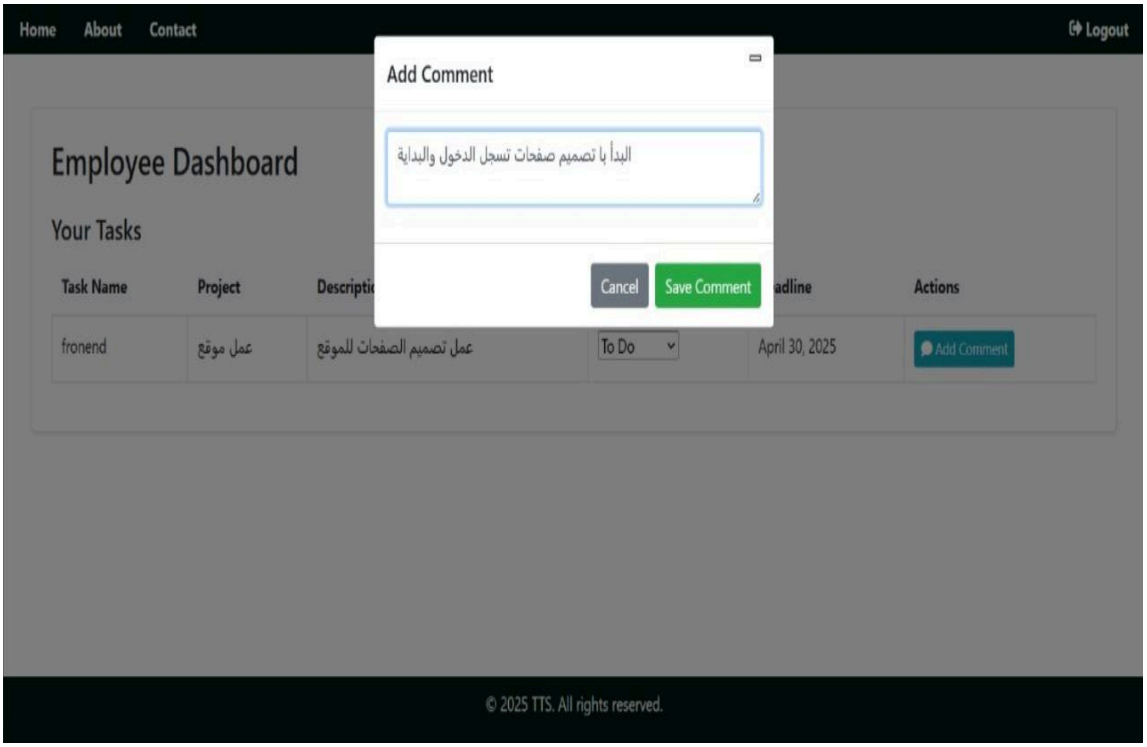


Figure 24 (Add Comment To Task Interface)

3.5.5. Add New Company Interface

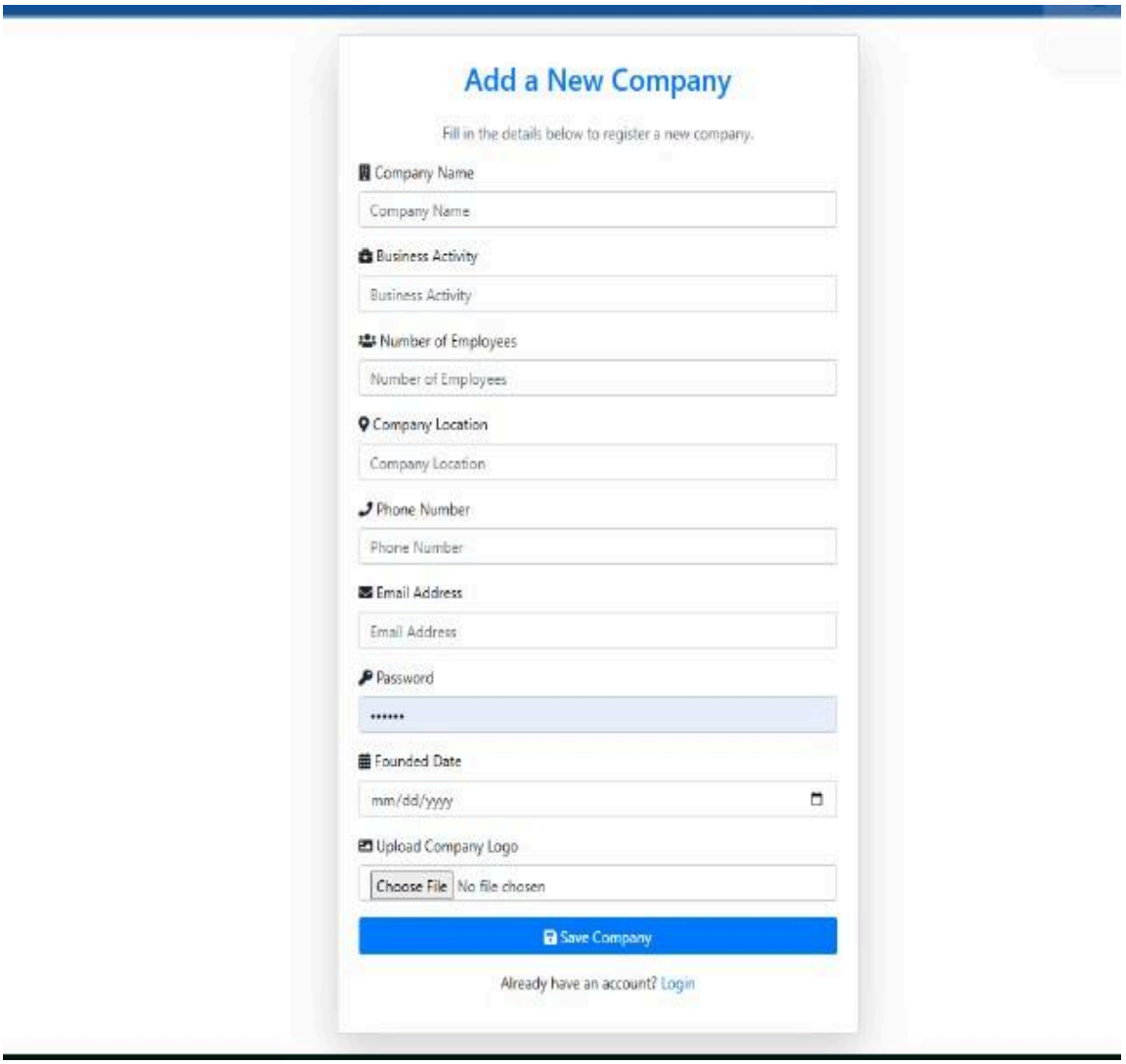


Figure 25 (Add New Company Interface)

3.6. General Conclusion

This chapter highlighted the critical decisions and tools used in designing and implementing the **Task Tracking System** (**TTS).** The chosen technologies, particularly Django and a relational database, align with the system's goals of security, scalability, and efficiency. The user interface was carefully crafted to meet usability standards and accommodate different user roles.

3.7. Future Work

While the current implementation of the **Task Tracking System** (**TTS**) covers the essential functionalities for managing tasks and user roles, there are several potential areas for improvement and expansion in future versions of the system:

- **Mobile Application**: Develop a dedicated mobile app version to enhance accessibility, especially for team members who need to update or track tasks on the go.
- **AI-Based Task Suggestions**: Integrate machine learning algorithms to analyze task history and recommend optimal task assignments, estimate completion times, and highlight project risks.

References

[1] “ Trello,” [Online]. Available: https://softyfile.com/trello/. [Accessed 20 09 2024].

“ Trello,” [Online]. Available: https://trello.com/u/saad53233592/boards. [Accessed 20 09 2024].

[2] “Monday.com,” [Online]. Available: https://www.laimuna.com/product/monday-com/. [Accessed 21 09 2024].

“Monday.com,” [Online]. Available: https://rabeea-team.monday.com/. [Accessed 21 09 2024].

[3] “ClickUp,” [Online]. Available: https://clickup.com/features. [Accessed 22 09 2024].

“ClickUp,” [Online]. Available: https://clickup.com/. [Accessed 22 09 2024].

[4] “Basecamp,” [Online]. Available: https://www.laimuna.com/product/basecamp/. [Accessed 23 09 2024].

“Basecamp,” [Online]. Available: https://basecamp.com/. [Accessed 23 09 2024].

[5] “Jira,” [Online]. Available: https://appmaster.io/ar/blog/m-hw-jyr. [Accessed 24 09 2024].

“Jira,” [Online]. Available: https://www.atlassian.com/software/jira. [Accessed 24 09 2024].

[6] “Scrum Master,” [Online]. Available: https://www.atlassian.com/agile/scrum/roles. [Accessed 29 09 2024].

“Scrum Master,” [Online]. Available: https://en.wikipedia.org/wiki/Agile_software_development. [Accessed 29 09 2024].

[7] “System Use Case,” [Online]. Available: https://app.diagrams.net/. [Accessed 01 11 2024].

[8] “System Use Case For Register as Project Manager,” [Online]. Available: https://app.creately.com/d/POJeWWcsm7a/edit. [Accessed 24 11 2024].

[9] “Sequence Diagram for Register as Project Manager,” [Online]. Available: https://app.creately.com/d/POJeWWcsm7a/edit. [Accessed 24 11 2024].

[10] “System Use Case For Authentication as Employee,” [Online]. Available: https://app.creately.com/d/POJeWWcsm7a/edit. [Accessed 24 11 2024].

[11] “Sequence Diagram for Authentication as Employee,” [Online]. Available: https://app.creately.com/d/POJeWWcsm7a/edit. [Accessed 24 11 2024].

[12] “System Use Case For Manage Task as Project Manager,” [Online]. Available: https://app.creately.com/d/POJeWWcsm7a/edit. [Accessed 24 11 2024].

[13] “Sequence Diagram for Manage Task as Project Manager,” [Online]. Available: https://app.creately.com/d/POJeWWcsm7a/edit. [Accessed 24 11 2024].

[14] “Class Diagram,” [Online]. Available: https://app.diagrams.net/. [Accessed 24 11 2024].